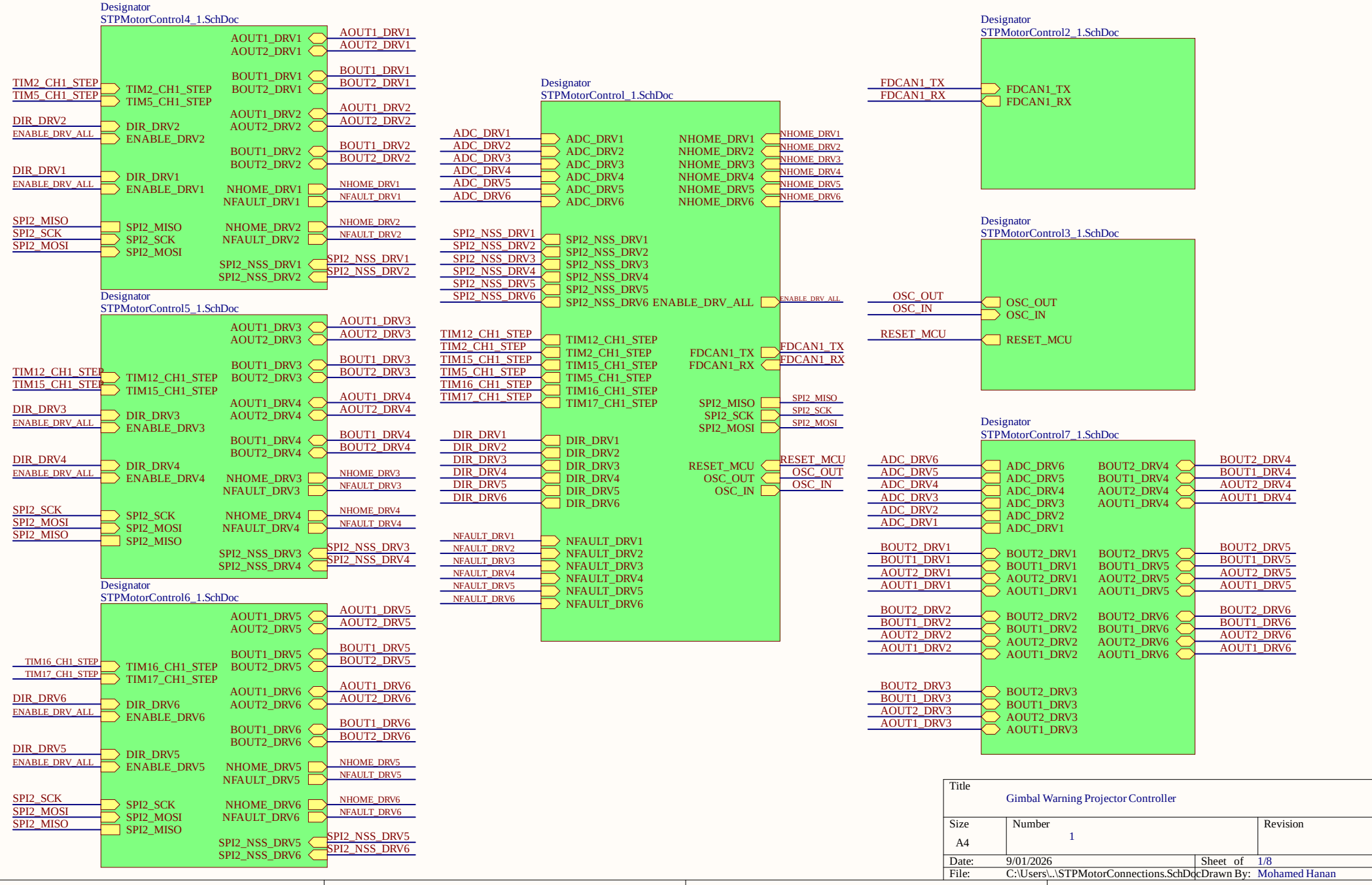
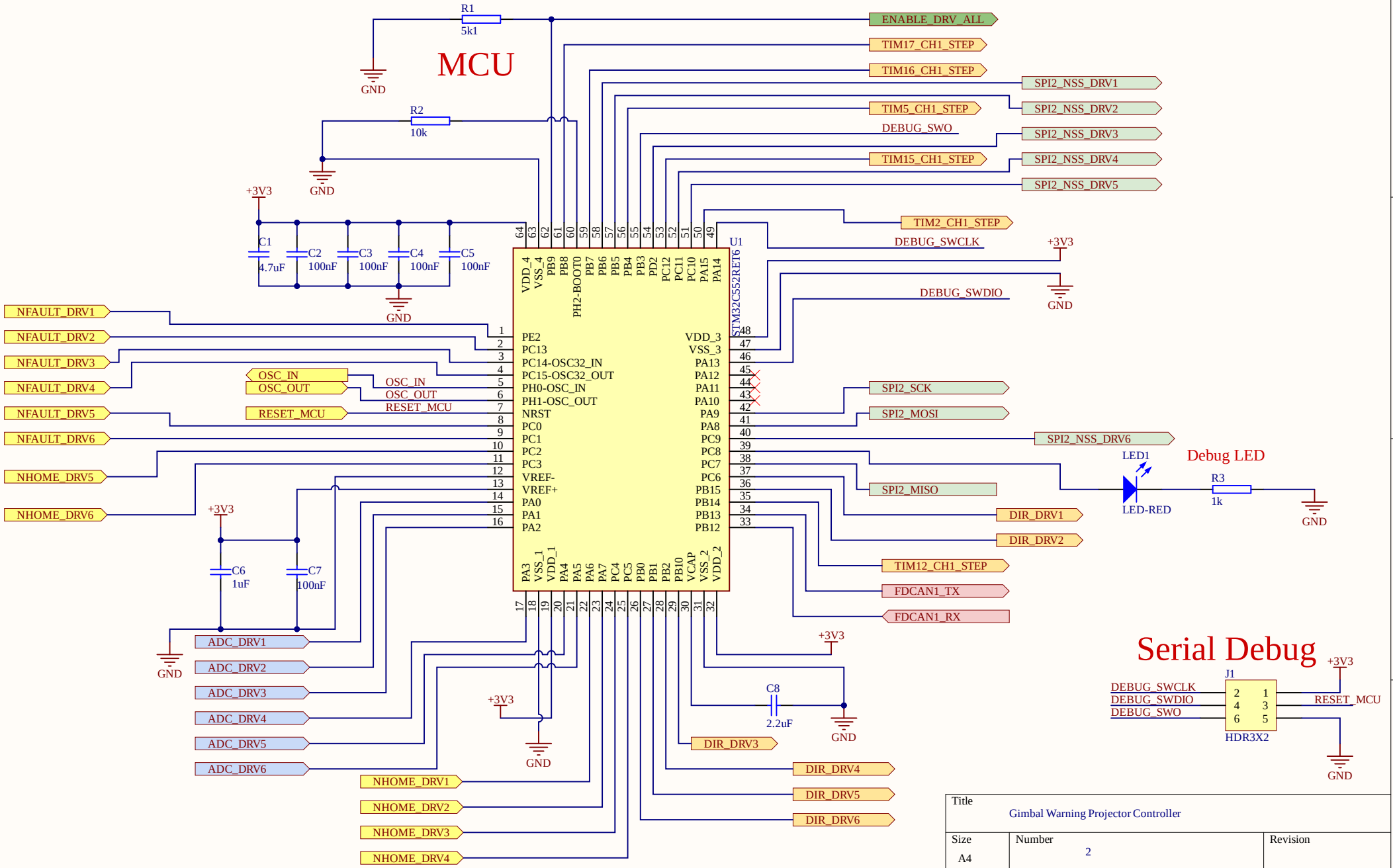
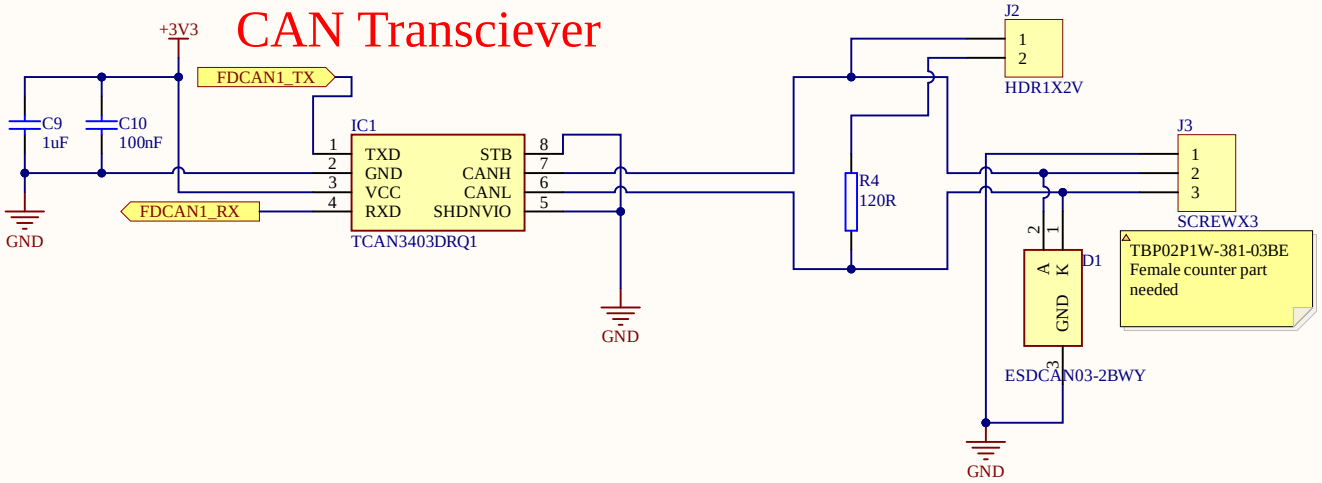

Designator
STPMotorControl4_1.SchDoc
AOUT1_DRV1
AOUT2_DRV1
BOUT1_DRV1
BOUT2_DRV1
AOUT1_DRV2
AOUT2_DRV2
BOUT1_DRV2
BOUT2_DRV2
NHOME_DRV1
NFAULT_DRV1
NHOME_DRV2
NFAULT_DRV2
SPI2_NSS_DRV1
SPI2_NSS_DRV2
TIM2_CH1_STEP
TIM5_CH1_STEP
DIR_DRV2
ENABLE_DRV_ALL
DIR_DRV1
SPI2_MISO
SPI2_SCK
SPI2_MOSI
Designator
STPMotorControl5_1.SchDoc
AOUT1_DRV3
AOUT2_DRV3
BOUT1_DRV3
BOUT2_DRV3
AOUT1_DRV4
AOUT2_DRV4
BOUT1_DRV4
BOUT2_DRV4
NHOME_DRV3
NFAULT_DRV3
NHOME_DRV4
NFAULT_DRV4
SPI2_NSS_DRV3
SPI2_NSS_DRV4
TIM12_CH1_STEP
TIM15_CH1_STEP
DIR_DRV3
ENABLE_DRV_ALL
DIR_DRV4
SPI2_SCK
SPI2_MOSI
SPI2_MISO
Designator
STPMotorControl6_1.SchDoc
AOUT1_DRV5
AOUT2_DRV5
BOUT1_DRV5
BOUT2_DRV5
AOUT1_DRV6
AOUT2_DRV6
BOUT1_DRV6
BOUT2_DRV6
NHOME_DRV5
NFAULT_DRV5
NHOME_DRV6
NFAULT_DRV6
SPI2_NSS_DRV5
SPI2_NSS_DRV6
TIM16_CH1_STEP
TIM17_CH1_STEP
DIR_DRV6
ENABLE_DRV_ALL
DIR_DRV5
SPI2_SCK
SPI2_MOSI
SPI2_MISO
Designator
STPMotorControl_1.SchDoc
ADC_DRV1
ADC_DRV2
ADC_DRV3
ADC_DRV4
ADC_DRV5
ADC_DRV6
NHOME_DRV1
NHOME_DRV2
NHOME_DRV3
NHOME_DRV4
NHOME_DRV5
NHOME_DRV6
SPI2_NSS_DRV1
SPI2_NSS_DRV2
SPI2_NSS_DRV3
SPI2_NSS_DRV4
SPI2_NSS_DRV5
SPI2_NSS_DRV6
ENABLE_DRV_ALL
TIM12_CH1_STEP
TIM2_CH1_STEP
TIM15_CH1_STEP
TIM5_CH1_STEP
TIM16_CH1_STEP
TIM17_CH1_STEP
FDCAN1_TX
FDCAN1_RX
SPI2_MISO
SPI2_SCK
SPI2_MOSI
DIR_DRV1
DIR_DRV2
DIR_DRV3
DIR_DRV4
DIR_DRV5
DIR_DRV6
RESET_MCU
OSC_OUT
OSC_IN
NFAULT_DRV1
NFAULT_DRV2
NFAULT_DRV3
NFAULT_DRV4
NFAULT_DRV5
NFAULT_DRV6
Designator
STPMotorControl2_1.SchDoc
FDCAN1_TX
FDCAN1_RX
Designator
STPMotorControl3_1.SchDoc
OSC_OUT
OSC_IN
RESET_MCU
Designator
STPMotorControl7_1.SchDoc
ADC_DRV6
ADC_DRV5
ADC_DRV4
ADC_DRV3
ADC_DRV2
ADC_DRV1
BOUT2_DRV4
BOUT1_DRV4
AOUT2_DRV4
AOUT1_DRV4
BOUT2_DRV1
BOUT1_DRV1
AOUT2_DRV1
AOUT1_DRV1
BOUT2_DRV5
BOUT1_DRV5
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AOUT1_DRV5
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AOUT1_DRV2
BOUT2_DRV6
BOUT1_DRV6
AOUT2_DRV6
AOUT1_DRV6
BOUT2_DRV3
BOUT1_DRV3
AOUT2_DRV3
AOUT1_DRV3
Title
Gimbal Warning Projector Controller
Size
A4
Number
1
Revision
Date: 9/01/2026
Sheet of 1/8
File: C:\Users\..\STPMotorConnections.SchDoc
Drawn By: Mohamed Hanan



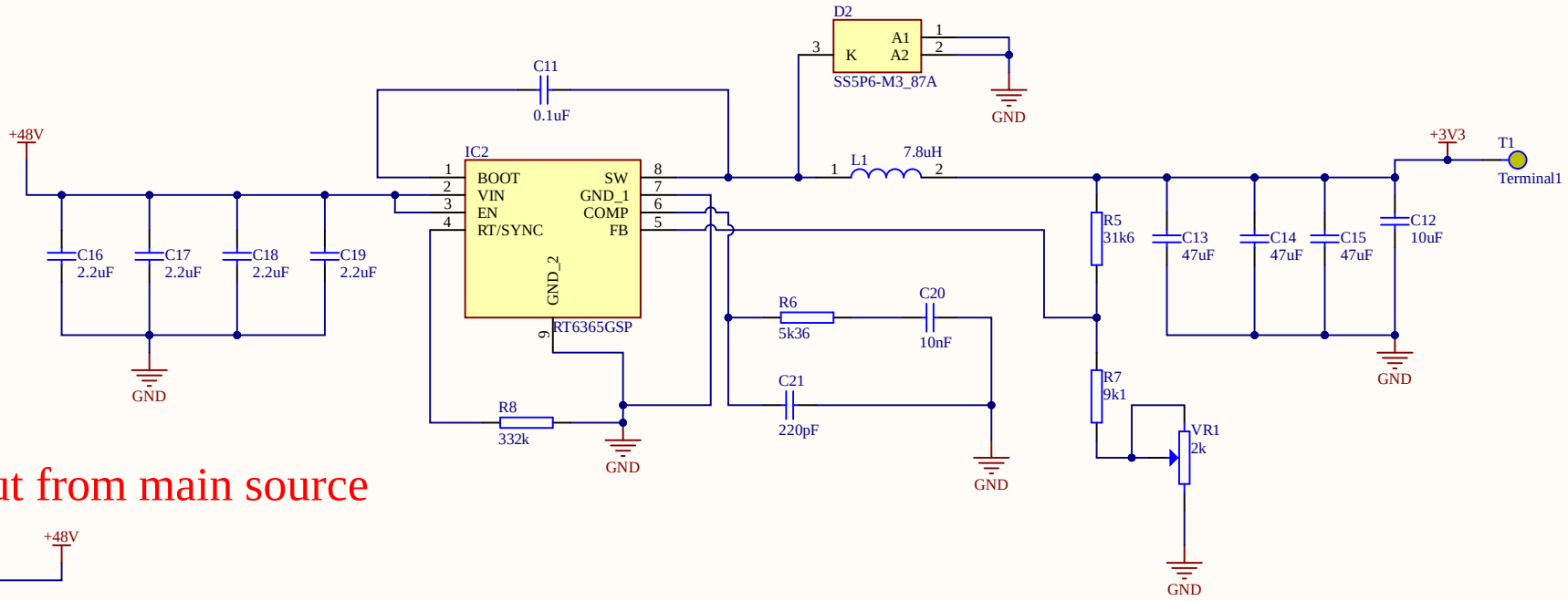
Title	Gimbal Warning Projector Controller	
Size: A4	Number: 2	Revision:
Date: 9/01/2026		Sheet of 2/8
File: C:\Users\..\STPMotorControl_1.SchDoc		Drawn By: Mohamed Hanan

CAN Transciever

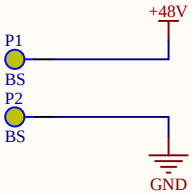


Title	Gimbal Warning Projector Controller		
Size: A4	Number: 3		Revision:
Date: 9/01/2026		Sheet of 3/8	
File: C:\Users\..\STPMotorControl2_1.SchDoc		Drawn By: Mohamed Hanan	

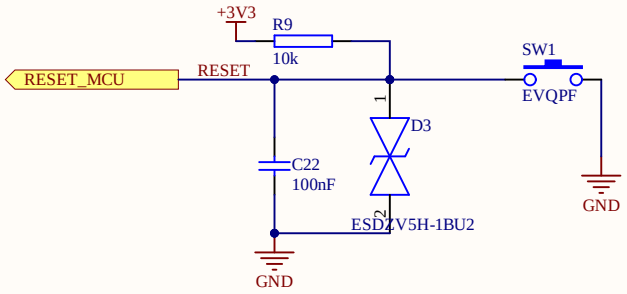
Switch mode converter form 48V to 3.3V



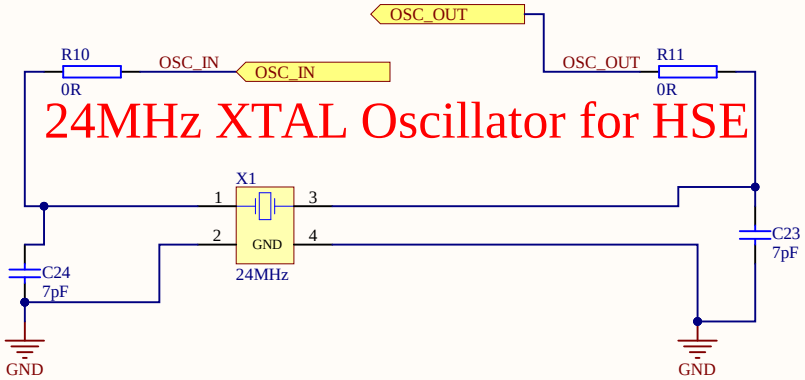
48V input from main source



Switch 4 MCU RESET PIN

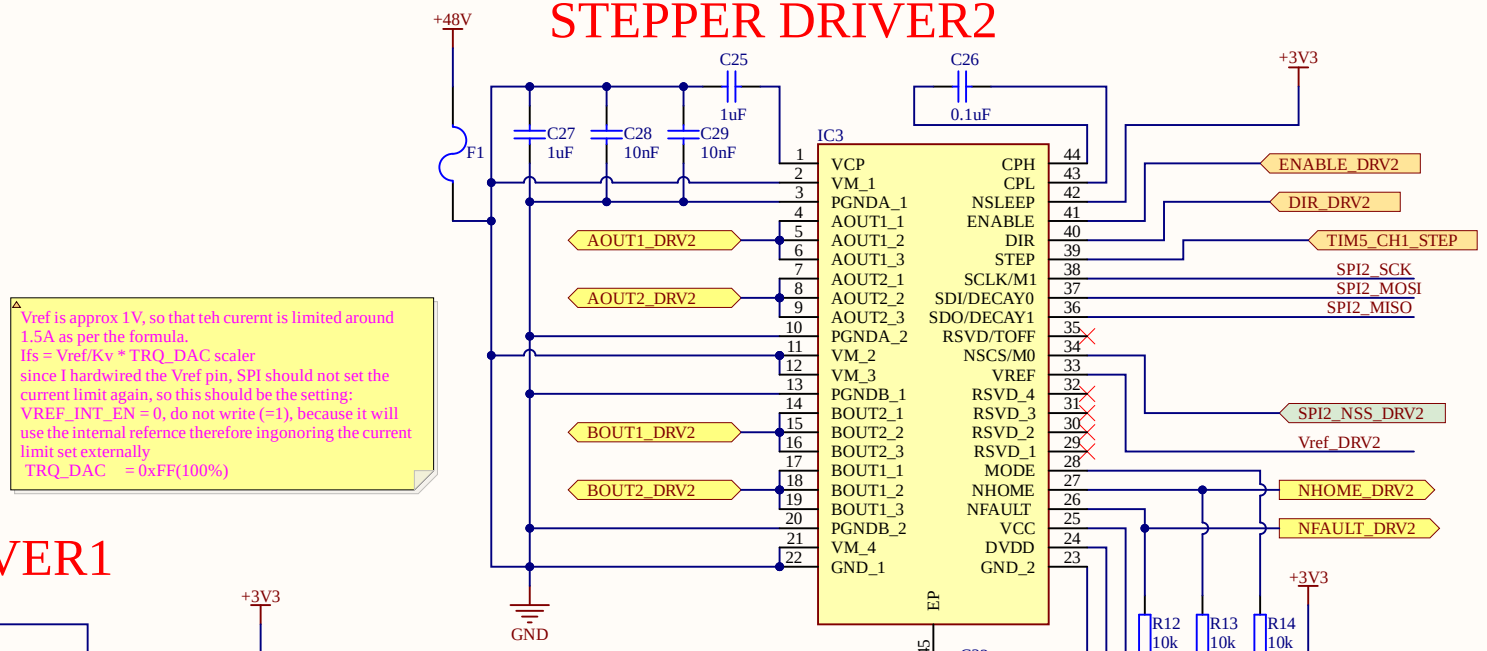


24MHz XTAL Oscillator for HSE



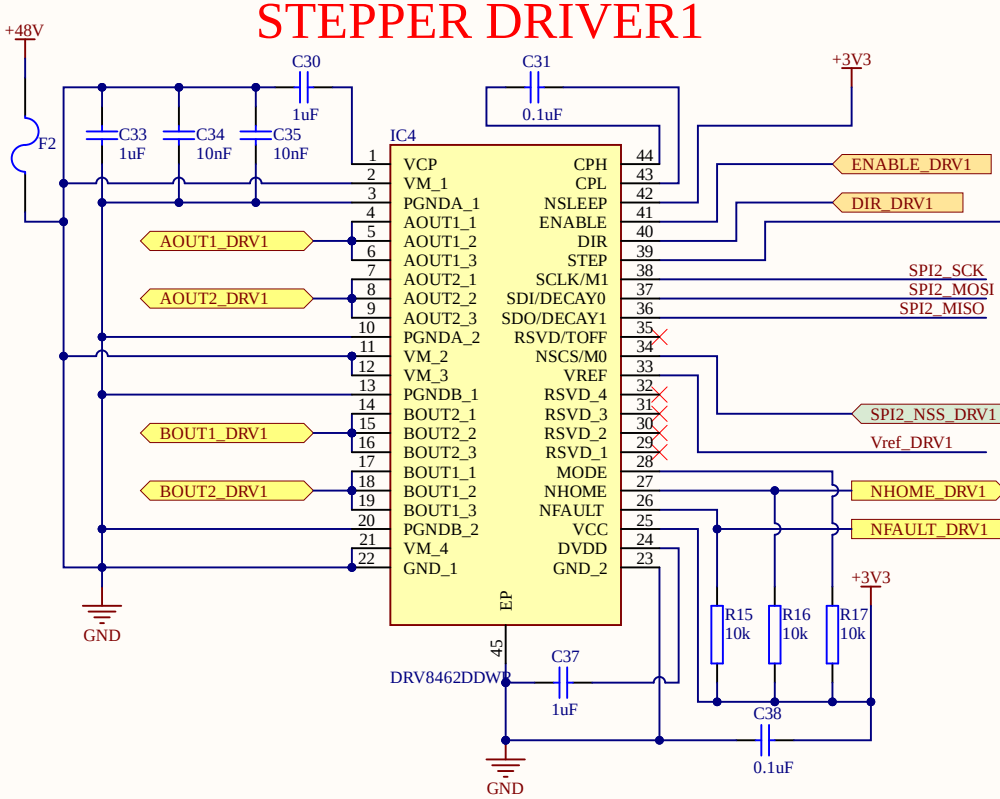
Title	Gimbal Warning Projector Controller		
Size: A4	Number: 4		Revision:
Date: 9/01/2026		Sheet of: 4/8	
File: C:\Users\..\STPMotorControl3_1.SchDoc		Drawn By: Mohamed Hanan	

STEPPER DRIVER2

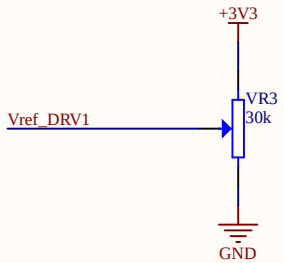


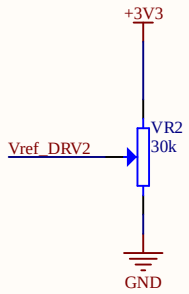
Vref is approx 1V, so that teh curernt is limited around 1.5A as per the formula.
Ifs = Vref/Kv * TRQ_DAC scaler
since I hardwired the Vref pin, SPI should not set the current limit again, so this should be the setting:
VREF_INT_EN = 0, do not write (=1), because it will use the internal refernce therefore ingonoring the current limit set externally
TRQ_DAC = 0xFF(100%)

STEPPER DRIVER1



Vref = 1V
so with 30k POT
3.3(9.1/(9.1+20.9)) = 1.001
1.001/0.66 = 1.51A

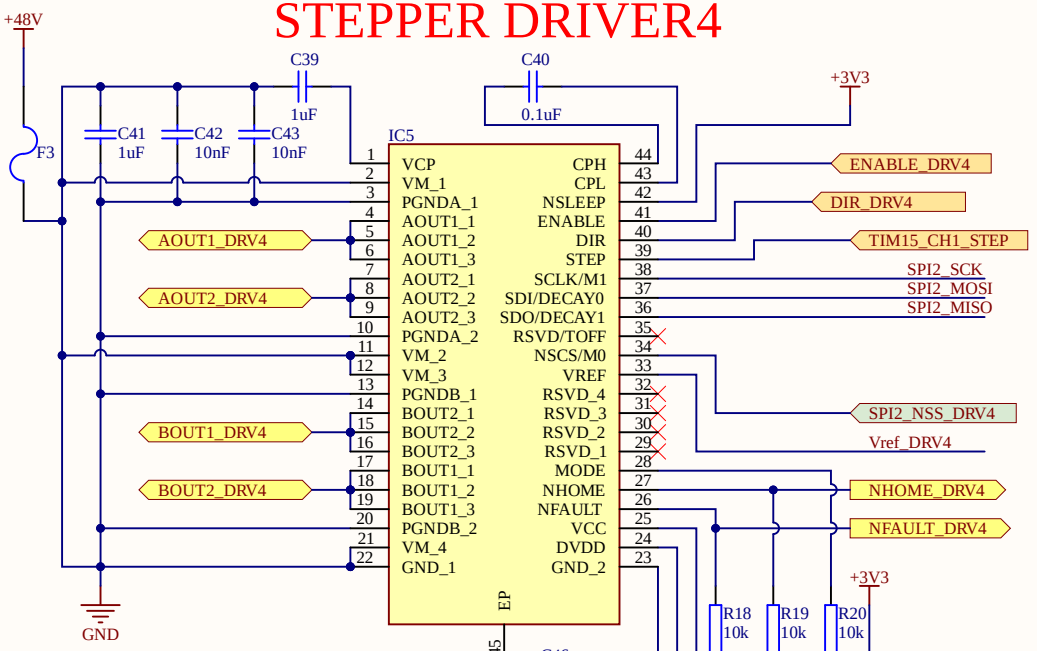




SPI2_SCK
SPI2_MOSI
SPI2_MISO

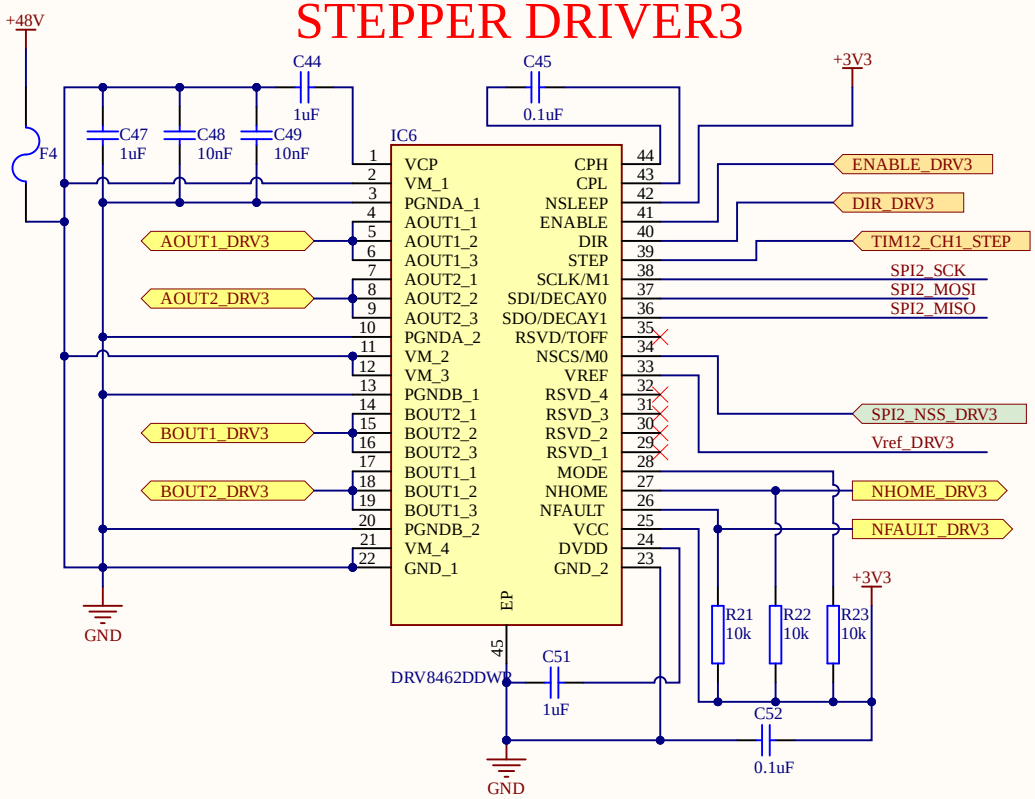
Title: Gimbal Warning Projector Controller		
Size: A4	Number: 5	Revision:
Date: 9/01/2026	Sheet of 5/8	
File: C:\Users\..\STPMotorControl4_1.SchDoc	Drawn By: Mohamed Hanan	

STEPPER DRIVER4

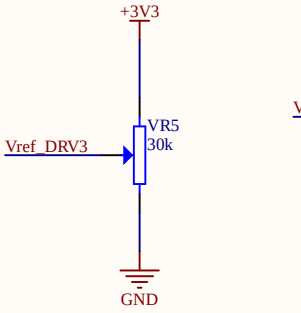


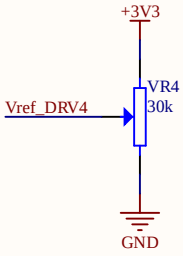
Vref is approx 1V, so that teh curernt is limited around 1.5A as per the formula.
Ifs = Vref/Kv * TRQ_DAC scaler
since I hardwired the Vref pin, SPI should not set the current limit again, so this should be the setting:
VREF_INT_EN = 0, do not write (=1), because it will use the internal refernce therefore ingonoring the current limit set externally
TRQ_DAC = 0xFF(100%)

STEPPER DRIVER3



Vref = 1V
so with 30k POT
3.3(9.1/(9.1+20.9)) = 1.001
1.001/0.66 = 1.51A

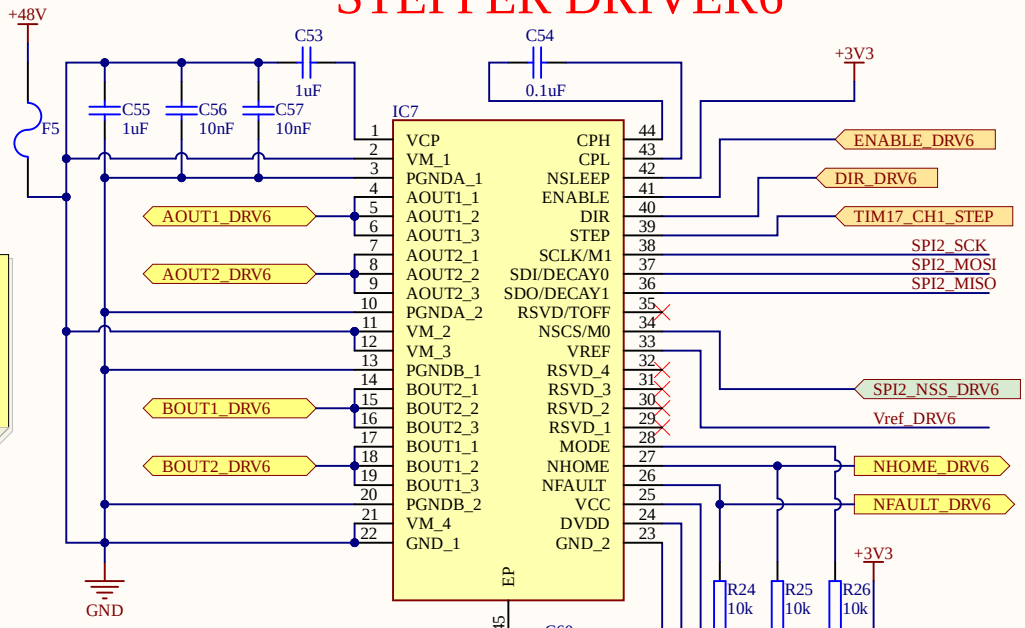




SPI2_SCK
SPI2_MOSI
SPI2_MISO

Title: Gimbal Warning Projector Controller		
Size: A4	Number: 6	Revision:
Date: 9/01/2026	Sheet of: 6/8	
File: C:\Users\..\STPMotorControl5_1.SchDoc	Drawn By: Mohamed Hanan	

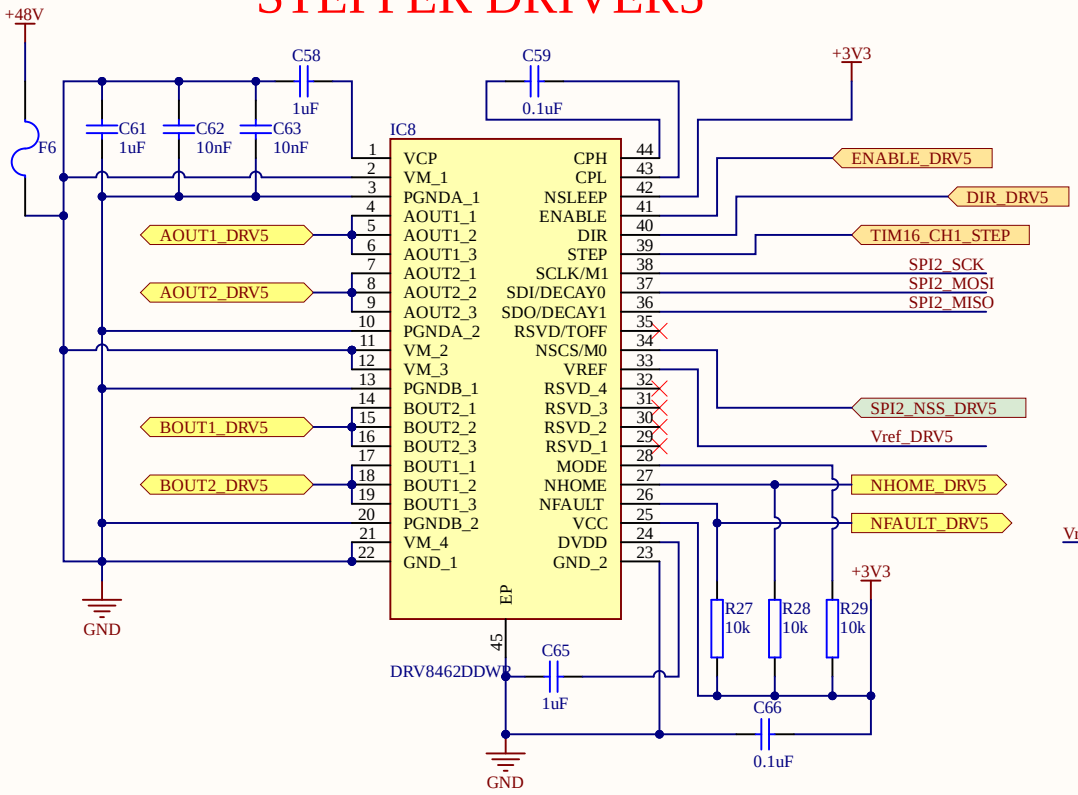
STEPPER DRIVER6

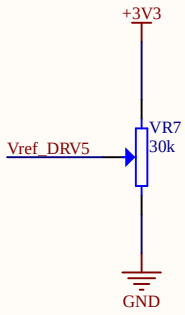


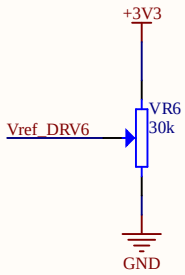
Vref is approx 1V, so that teh curernt is limited around 1.5A as per the formula.
Ifs = Vref/Kv * TRQ_DAC scaler
since I hardwired the Vref pin, SPI should not set the current limit again, so this should be the setting:
VREF_INT_EN = 0, do not write (=1), because it will use the internal refernce therefore ingonoring the current limit set externally
TRQ_DAC = 0xFF(100%)

Vref = 1V
so with 30k POT
3.3(9.1/(9.1+20.9)) = 1.001
1.001/0.66 = 1.51A

STEPPER DRIVER5



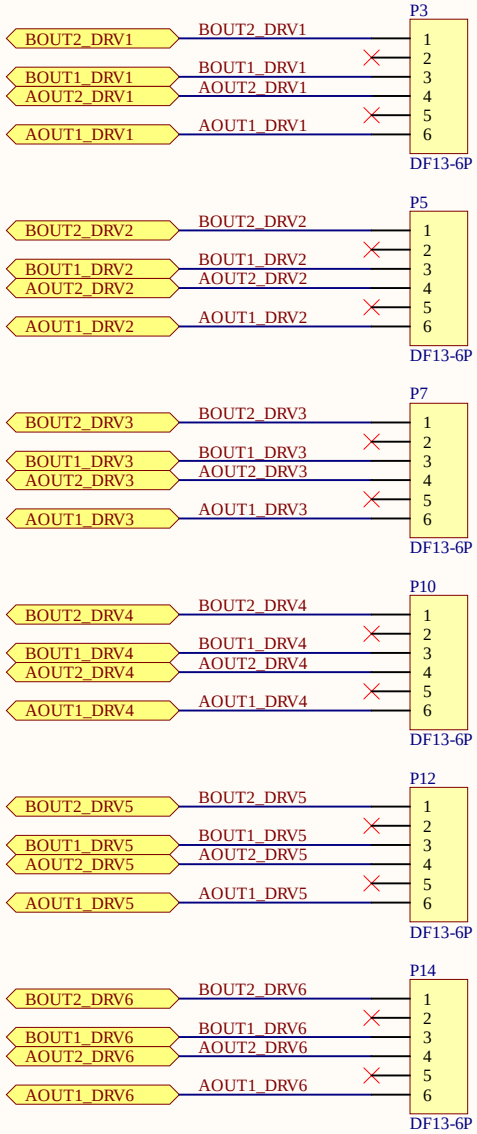




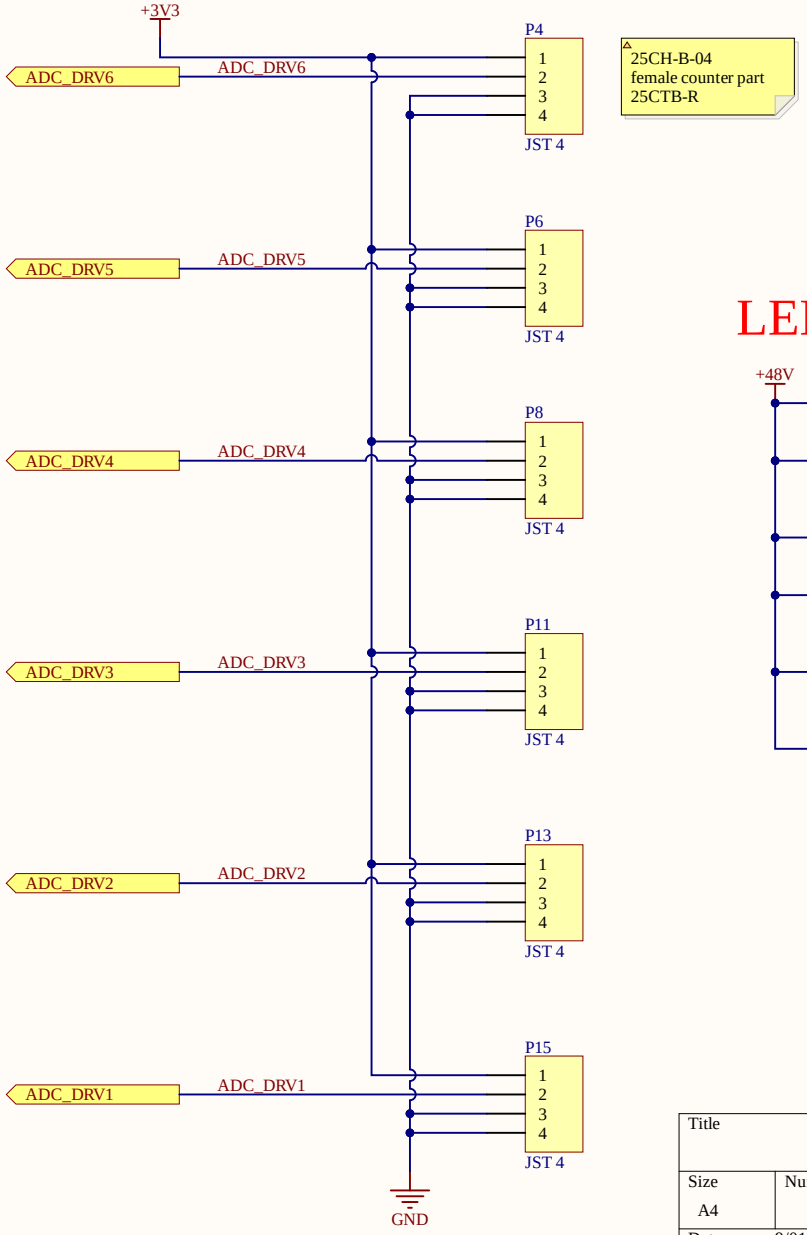
SPI2_SCK
SPI2_MOSI
SPI2_MISO

Title: Gimbal Warning Projector Controller		
Size: A4	Number: 7	Revision:
Date: 9/01/2026	Sheet of: 7/8	
File: C:\Users\..\STPMotorControl6_1.SchDoc	Drawn By: Mohamed Hanan	

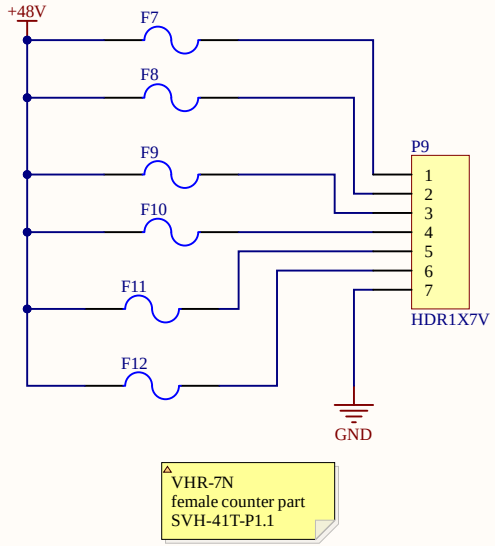
STEPPER MOTOR CONNECTOR



MAGNETIC ENCODER CONNECTOR



LED modules Connector



Title: Gimbal Warning Projector Controller		
Size: A4	Number: 8	Revision:
Date: 9/01/2026	Sheet of: 8/8	
File: C:\Users\..\STPMotorControl7_1.SchDoc	Drawn By: Mohamed Hanan	